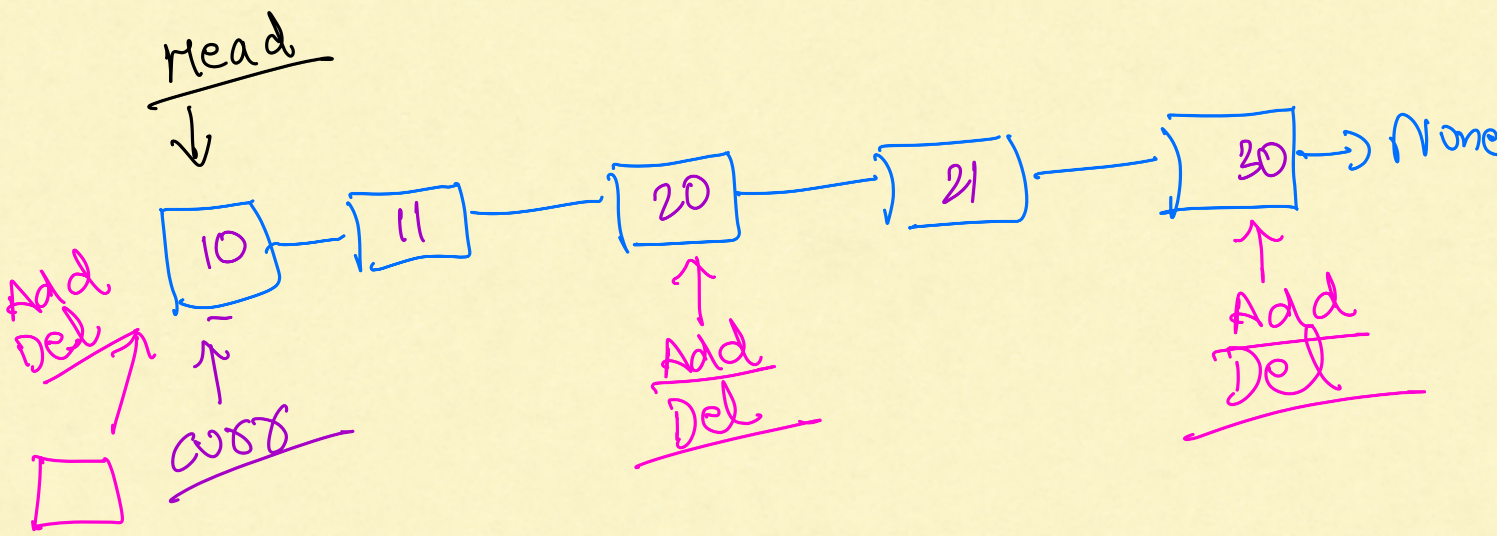
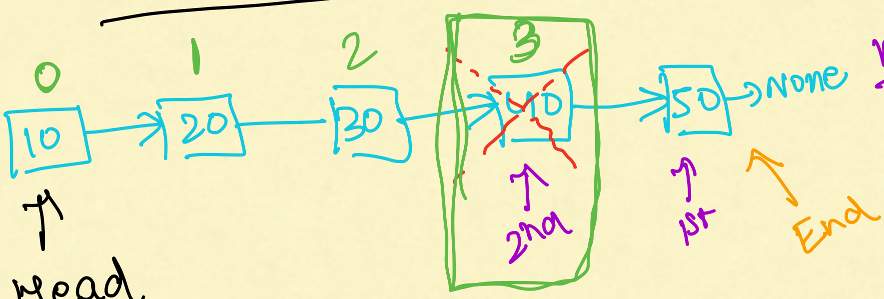
LINKED LIST


imgflip.com

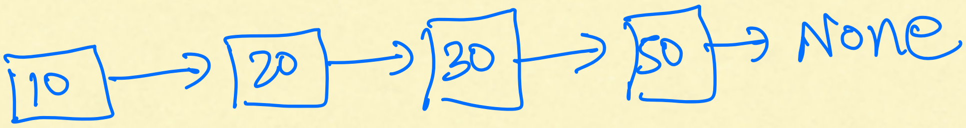

head
10
11
20
21
30
None
Add
Del
curr
Add
Del
Add
Del

Remove N^{th} Node from the End of LL



n=2

count = 5

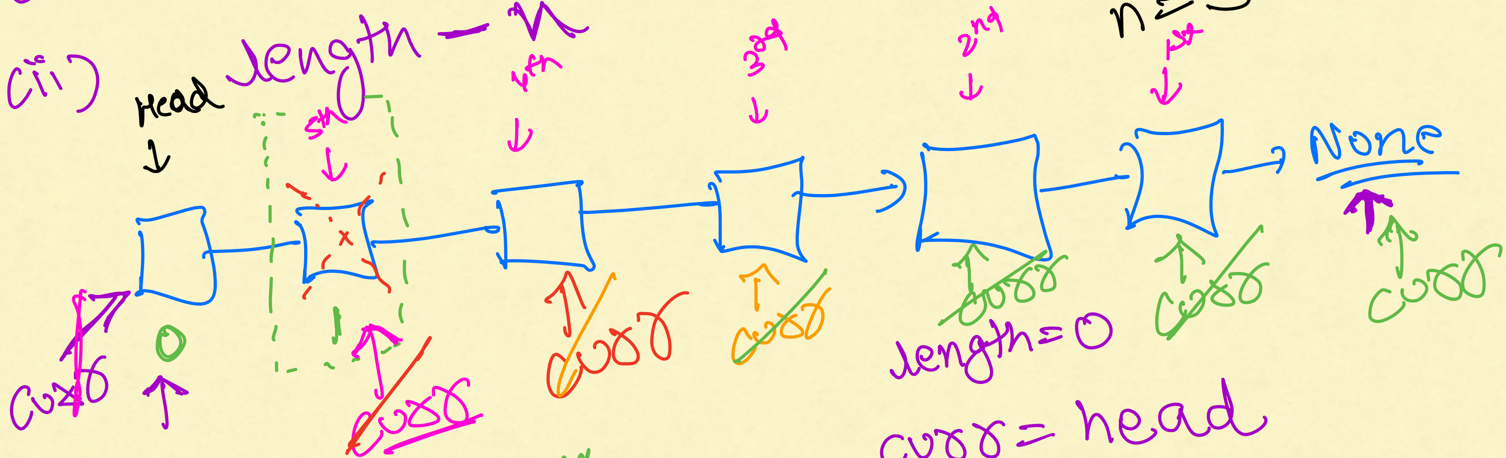


count = 5 - n
= 5 - 2
= 3

(i) cal the length of LL

(ii) length - n

n = 5

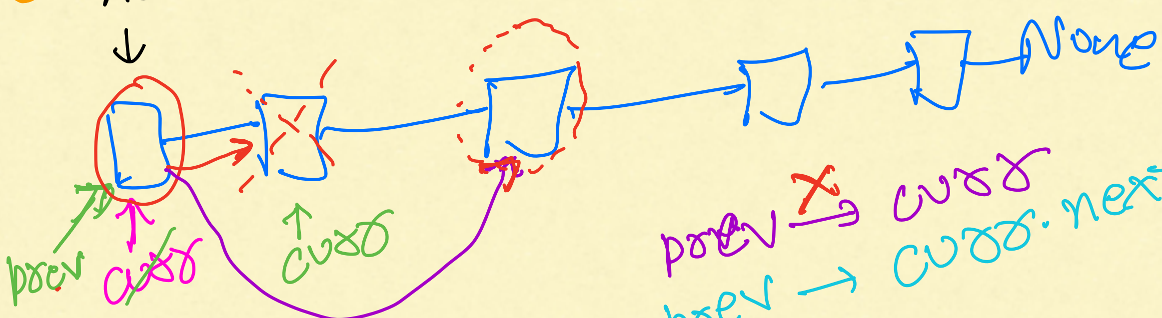


length = ~~0~~ ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 6

length = 0
curr = head
while curr != None:
    length = length + 1
    curr = curr.next

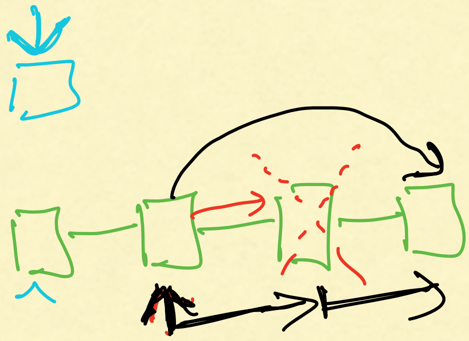
None != None

length = 6



prev ~~→~~ curr
prev → curr.next

curr = head
count = 1
while (count < (length - n)):
count += 1
curr = curr.next



curr.next = curr.next.next

Pro Tip
NEVER change head ⇒ curr
temp
rahul

arr = [11, 5, 3, 1, 7]
~~arr.sort()~~
⇒ [1, 3, 5, 7, 11]

ast term
PPE

Merge Sort (LL)

1 3 5 7 11

~~11~~ → ~~5~~ → ~~3~~ → ~~1~~ → ~~7~~ → None

⇓

1 → 3 → 5 → 7 → 11 → None

Merge two sorted LL

without extra space

prev H1 i prev i

L1: 1 → 3 → 5 → 7 → None

$i < j$

L2 2 → 4 → 6 → 8 → None

i

j H2 prev j

⇓

1 → 2 → 3 → 4 → 5 → 6 → 7 → 8 → None

copy paste array merge

1 →

temp_head

-1

temp

temp-head

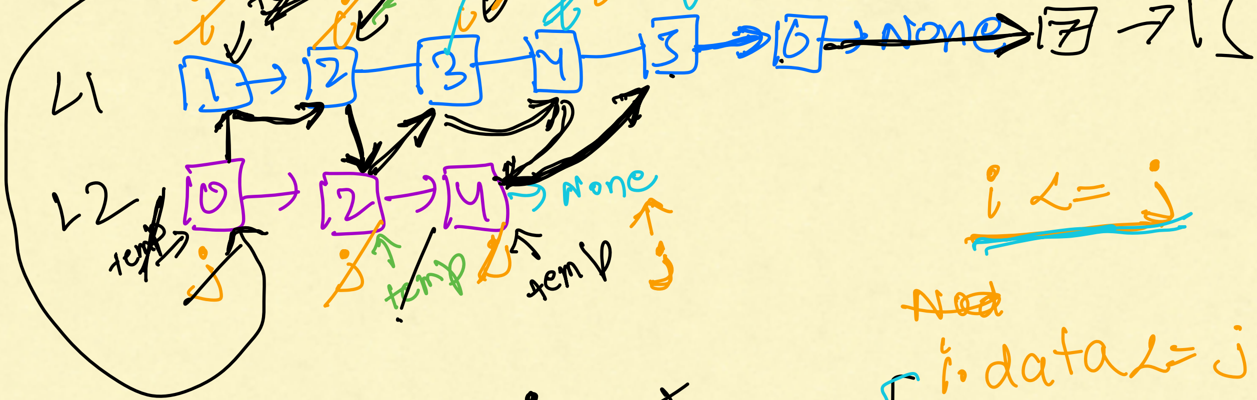
head

~~temp~~

-1 small

arr

~~temp~~ ~~temp~~ ~~temp~~ ~~temp~~ i



i <= j

~~Head~~

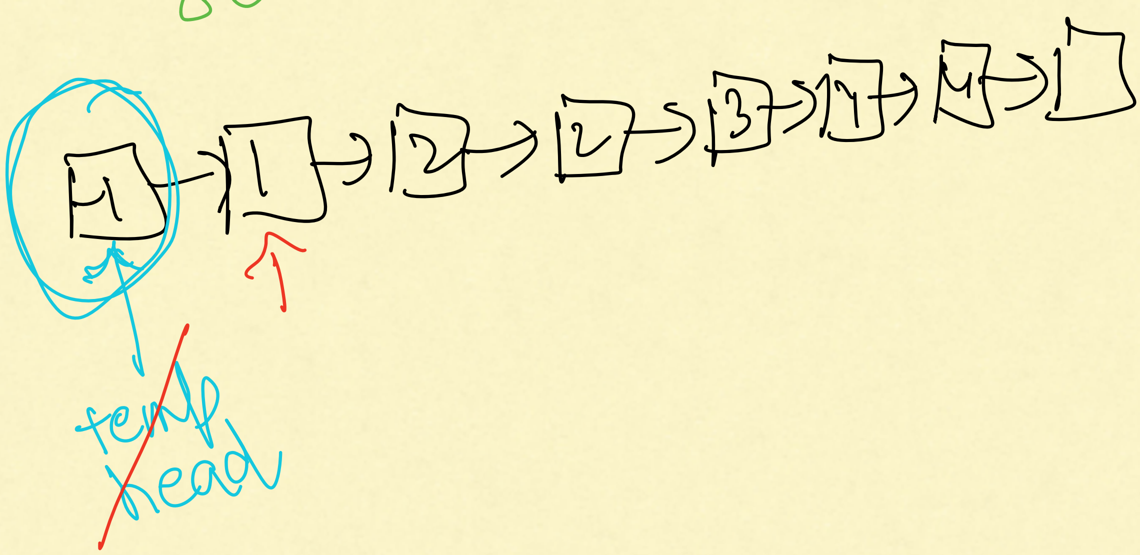
i.data <= j.data
temp.next = i
temp = temp.next
i = i.next

temp.next = j
temp = temp.next
j = j.next

while (i != None):
    temp.next = i
    break
while (j != None):
    temp.next = j
    break

return ~~temp_head~~

temp_head.next



TRICK

If u don't known how to solve the LL question

⇓

convert it into array(list)/set/dict, do all modification inside your array/set/dict

⇓

paste the value inside LL.

ex: ① Sort LL

[3] → [1] → [6] → [5] → None

⇓

arr = [3, 1, 6, 5]

arr.sort ⇒ [1, 3, 5, 6]

1 3 5 6

[~~3~~] → [~~1~~] → [~~6~~] → [~~5~~] → None

ex: ② remove duplicate from unsorted LL

[5] → [2] → [2] → [2] → [2] → [4] → None

⇓ put the no. side set

set = {5, 2, 4}

5 2 4 None

[~~5~~] → [~~2~~] → [~~2~~] → [2] → [2] → [4] → None

ex: ③ Remove N^{th} Node From End of List

k=2

[1] – [2] – [3] – [~~4~~] → [5] → None

⇓

arr = [1, 2, 3, 4, 5]

⇓ reverse the arr (arr. reverse)

arr = [5, ~~4~~, 3, 2, 1]

Now delete K^{th} no. from arry

copy past

before that reverse

[1, 2, 3, 5]

⇑

1 2 3 5. None

[~~1~~] → [~~2~~] – [~~3~~] – [~~4~~] → [5]